

Is the twelve-rung HAR ladder three-dimensional?

Method note: supervised rank-3 compression against the full ladder

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Abstract

We ask whether the twelve rungs of a base-2 HAR ladder carry twelve degrees of freedom or fewer. Every arm keeps the same twelve lags $\{1, 2, 4, \dots, 2048\}$ and differs only in how many shape coefficients the ladder is allowed. Under fixed *a priori* bases the ladder does not compress. Under a supervised basis it does: three coefficients in twelve columns beat all twelve rungs in twenty-one columns by 0.00059 QLIKE ($t = -3.30$), with an interior optimum at $r = 3$. Principal components at the same rank are the worst arm tested. The result is not yet robustness-tested and the projection is estimated from a single frozen window; §8 states what must be conceded.

1 Panel and rows

Cache `results/b2_mmap_warm`, the corrected-fill build: availability recorded *before* the forward fill, a 26-bar staleness bound, scale estimated from observed rows only, and a warm-up guard on feed initiation.

- 242,934 bars on a 30-minute grid, Sunday 18:30 through Friday 20:00, 2005-01-02 to 2024-04-30.
- Training window $W = 24,000$ bars; the walk begins at bar 24,000.
- 218,934 predictions per arm; 218,909 scored after dropping bars whose truth is non-positive or whose features are non-finite.
- The scored row set is *identical* across arms, so every comparison is bar-for-bar.

2 Target

`robust_transform(df, "RV")` chains

diurnal adjustment $\longrightarrow$ `sqrt` semantic transform $\longrightarrow$ rolling winsorize

The diurnal step is multiplicative and returns the baseline b . Winsorization uses a 240-bar rolling window at the **1/99** quantiles—the interior optimum measured under a fixed truth, where the project default of 5/95 costs 0.00304 and removing the clip entirely does not help. The model trains on the transformed target y ; raw variance is reconstructed as $y^2 b$.

Scoring is on the **clipped 5/95** footing, $y_{5/95}^2 b$, which is the footing the paper’s battery uses and where its OLS HAR base-5 plus calendar sits at 0.13415.

3 Features

Let $s = y^2$ be the bar-level adjusted variance. For each rung $L \in \{1, 2, 4, \dots, 2048\}$,

```
H[:, k] = sqrt( s.rolling(L, min_periods=L).mean().shift(1) )
```

The `.shift(1)` is the causality guard: bar t 's feature uses data only through $t-1$. Nine calendar columns are appended—`DOW_0..4`, `hour`, `is_overnight`, `is_open`, `is_close`. **No exogenous channels are used.**

4 The two arms

Both designs are $[HG, \text{cal}]$ with an intercept; only G differs.

Baseline, base-2 (r=12). $G = I_{12}$, giving $12 + 9 = 21$ columns plus intercept—the untouched ladder.

Compressed, sup r=3. $G \in \mathbb{R}^{12 \times 3}$, giving $3 + 9 = 12$ columns plus intercept. G is built on rows $[0, W)$ *only*:

```
C = cov(H[:W])                # 12 x 12 ladder covariance
xy = (H[:W]-mean).T @ (y[:W]-mean)/W # ladder-target cross-covariance
d0 = solve(C, xy)              # the OLS coefficient direction
fam = [d0, C@d0, C@C@d0, ...]  # Krylov space K_r(C, d0)
G = gram_schmidt(fam)[:r]
```

This is a **Krylov / PLS-type subspace**, not a generic “supervised projection”, and the distinction matters for how the result should be described. At $r = 1$ it is the frozen OLS coefficient direction. Because $Cd_0 = xy$, for $r \geq 2$ the span contains both the OLS direction and the raw cross-covariance direction.

5 Estimator: per-bar OLS, exact

Both arms refit at *every bar*, with no penalty, matching the battery’s convention rather than this project’s usual 1,000-bar cadence. A window step changes $X^\top X$ by rank two—one row leaves, one enters—so the inverse is updated in $O(p^2)$ rather than re-solved in $O(p^3)$:

```
cf = Ainv @ c
u = A[t-W]; v = A[t]                # row leaving, row entering
Ainv += outer(Au,Au) / (1 - u.T@Ainv@u) # remove
Ainv -= outer(Av,Av) / (1 + v.T@Ainv@v) # add
c = c - u*y[t-W] + v*y[t]
```

At $p \approx 20$ this is roughly 500 flops per bar, so all 218,934 refits cost less than a single 1,000-bar walk of a wide design. The fastest method here is also the correct protocol, which is why it is worth the care.

Rank-2 updating accumulates floating-point error, so the inverse is recomputed exactly every 5,000 bars and the worst observed relative drift is reported: 1.3×10^{-12} for **sup**, 1.0×10^{-13} for the baseline. The fast path is exact, not approximately exact.

6 Prediction and scoring

$$\widehat{RV}_t = \left((A_t \hat{\beta})^2 + \frac{1}{W} \sum_i r_i^2 \right) b_t \quad (\text{Duan smearing})$$

$$\text{QLIKE} = \rho - \log \rho - 1, \quad \rho = RV_t / \widehat{RV}_t$$

Inference is Diebold–Mariano with a Newey–West HAC variance at the rule-of-thumb bandwidth and the Harvey–Leybourne–Newbold small-sample correction. Eras are eight equal blocks of the scored index.

7 Result, and its retraction

arm	lag dims	cols	QLIKE	vs base-2
base-2 (r=12)	12	21	0.13438	—
sup r=3	3	12	0.13379	−0.00059 ($t = -3.30$)

Three coefficients appear to beat twelve rungs, with an *interior* optimum rather than a monotone preference for fewer columns: -0.00017 , -0.00024 , **-0.00059** at $r = 3$, then $+0.00031$ and $+0.00032$.

7.1 Refitting the basis destroys it

The projection above is estimated once on rows $[0, W)$ and applied through 2024. Rebuilding it causally from the trailing window at three cadences gives:

arm	bases	QLIKE	vs base-2	max principal angle
base-2 (r=12)	—	0.13438	—	—
sup r=3 frozen	1	0.13358	−0.00080	0.00°
sup r=3 refit/24000	10	0.13465	+0.00027	89.02°
sup r=3 refit/6000	37	0.13461	+0.00023	84.42°
sup r=3 refit/1000	219	0.13464	+0.00026	66.09°

Every refit cadence loses. Eras better than base-2: 5 of 8 frozen, against 1, 2 and 1 of 8 for the refit arms.

7.2 The instability is the estimator, and fixing it removes the win

The subspace is built as a Krylov space $K_r(C, \text{seed})$. Three seeds were compared, holding the construction fixed so any difference is attributable to the seed alone:

seed	arm	QLIKE	vs base-2	principal angles per direction
$C^{-1}xy$	frozen	0.13358	−0.00080	—
$C^{-1}xy$	refit/6000	0.13461	+0.00023	[2.2, 63.8, 84.4]°
xy (PLS)	frozen	0.13460	+0.00023	—
xy (PLS)	refit/6000	0.13470	+0.00032	[1.0, 37.4, 59.1]°
$(C + \lambda I)^{-1}xy$	frozen	0.13378	−0.00060	—
$(C + \lambda I)^{-1}xy$	refit/6000	0.13457	+0.00020	[1.9, 49.0, 77.6]°

Standard PLS does not reproduce the win even frozen ($+0.00023$ against -0.00080). The ridge-shrunk seed sits between the two at -0.00060 , so the effect scales with how much the collinear ladder covariance is inverted. It is a property of the estimator, not a three-dimensional structure in the data.

The per-direction angles say where the instability lives. The first Krylov direction is stable to 1–2°; the second and third swing by 37 to 84°. And the frozen curve runs -0.00017 , -0.00024 , then **-0.00080** at $r = 3$: the entire win rides on precisely the two directions that do not survive refitting.

Two corrections to an earlier draft of this note. It attributed the instability to an ill-conditioned inversion; $\text{cond}(C)$ is 370–410 on this panel, collinear but not singular, so that was wrong. And it reported a single angle of 89.0° as showing the subspace “nearly orthogonal” to itself. That figure is $\arccos$ of the *smallest* singular value, hence the *largest* principal angle: it described one direction of three and was silent about the other two.

The compression claim is **withdrawn**. What transfers is the diagnostic: check a projection for subspace stability, per direction, and re-seed it a second well-conditioned way before believing its loss. A mean loss cannot distinguish a stable feature of the data from a lucky freeze, and a single summary angle cannot distinguish a subspace that drifts from one whose leading direction is fine and whose tail is noise.

7.3 A second problem: implementation sensitivity

The frozen arm scores 0.13379 under one implementation and 0.13358 under another. The two differ only in how non-finite feature values are handled—zeroing individual entries against zeroing whole rows—and the gap, 0.00021, is a quarter of the effect being claimed. Any result of this size on this panel needs its NaN policy pinned before the number means anything.

For context, the fixed bases at $r = 3$ all lose: power +0.00128, log-polynomial +0.00108, and principal components +0.00135—the worst of the four. Unsupervised compression of the ladder is actively harmful, matching the exogenous block, where a supervised projection beat the 516-dimensional ambient view while principal components and Laplacian eigenmaps lost. Variance directions and prediction directions are different objects here, and only the second compresses.

8 What must be conceded

1. **The freeze was the result**, not a simplification of it. Every causal refit cadence loses to the full ladder. This was the largest assumption in the design and it did not hold.
2. **The win depends on the seed**, not on the rank. A well-conditioned PLS seed does not reproduce it, and a partially shrunk one reproduces it partially.
3. **The unstable directions are the ones carrying the win**: direction one is stable to $1\text{--}2^\circ$, directions two and three swing by $37\text{--}84^\circ$, and the gain appears only at $r = 3$.
4. **The frozen number moves by 0.00021 across implementations** that differ only in NaN handling—a quarter of the claimed effect.
5. **The t -statistics rest on the rule-of-thumb HAC bandwidth**, which has produced two false positives elsewhere in this project.
6. **This is not a reproduction of the battery**. Base-2 should land near 0.13333, being 0.00082 better than base-5’s 0.13415, and 0.13438 is 0.00105 the wrong way. The training clip is 1/99 rather than 5/95 and the panel differs. The agreement confirms the footing and nothing more.

Artifacts: `analysis/ladder_headline.py`, `writeup/stats/ladder_headline.json`.